

Fake News Detection Using Natural Language Processing and Deep Learning Techniques

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Abstract—Social media and online news portals allow information to spread to millions of users within minutes, but they also allow false and misleading news to spread just as fast. Fake news can influence public opinion, elections, health decisions and financial markets, and manual fact-checking cannot keep up with the volume of content being published. This paper presents an automatic fake news detection system based on Natural Language Processing (NLP). The system analyses the headline and body text of a news article and classifies it as real or fake. Three families of models are studied: classical machine learning classifiers on TF-IDF features, a hybrid CNN-LSTM deep learning model using pre-trained word embeddings, and a fine-tuned BERT transformer model. The paper describes publicly available benchmark datasets, the preprocessing pipeline, the architecture of each model and the evaluation strategy. The goal of the work is to build a reliable, explainable tool that can assist readers and fact-checkers in identifying suspicious news content at an early stage.

Keywords—Fake News Detection, Natural Language Processing, Misinformation, TF-IDF, CNN-LSTM, BERT, Text Classification

I. INTRODUCTION

News consumption has moved from newspapers and television to social media platforms and messaging applications. While this makes information easily accessible, it also makes it easy for anyone to publish content without editorial checks. Fake news is defined as news articles that are intentionally and verifiably false and could mislead readers [1]. A large-scale study of Twitter by Vosoughi et al. [2] found that false news spreads farther, faster and deeper than true news, which shows how serious the problem has become.

Professional fact-checking organisations verify claims manually, but this process is slow and does not scale to the huge amount of content published every day. Therefore, automatic detection methods are required. Since the content of a news article is written in natural language, NLP techniques can be used to learn the linguistic patterns, writing style and word usage that differ between fake and genuine news [4]. Fake articles often use emotional words, exaggerated headlines, fewer sources and a sensational tone, and these signals can be captured by machine learning and deep learning models.

The main objectives of this work are:

- To study existing approaches and benchmark datasets for fake news detection.
- To design an NLP preprocessing and feature extraction pipeline for news articles.

- To implement and compare classical machine learning, CNN-LSTM and BERT-based classifiers for real/fake news classification.
- To provide a simple web interface in which a user can paste a news article and receive a prediction with a confidence score.

Section II presents the literature survey, Section III explains the datasets and the proposed methodology, and Section IV describes the system architecture and evaluation plan.

II. LITERATURE SURVEY

Shu et al. [1] presented a detailed survey of fake news detection from a data mining perspective and divided the approaches into news-content-based and social-context-based methods. Wang [3] released the LIAR dataset containing about 12.8K short political statements labelled with six degrees of truthfulness, and showed that a hybrid CNN using text and metadata performs better than text-only models. Pérez-Rosas et al. [4] built datasets covering several news domains and studied linguistic features such as n-grams, punctuation, readability and psycholinguistic categories for fake news detection.

Ahmed et al. [5] used n-gram features with TF-IDF weighting and compared six classical classifiers, finding that linear models such as Linear SVM work well; their ISOT dataset is widely used in student and research projects. Shu et al. [6] introduced FakeNewsNet, which includes news content along with social context from PolitiFact and GossipCop. Ruchansky et al. [7] proposed CSI, a hybrid deep model that combines the text of articles, the response of users and the behaviour of sources.

Deep learning models remove the need for manual feature engineering. CNNs [8] capture local n-gram patterns and LSTMs [9] capture long-range dependencies in text. Transformer-based models such as BERT [10] learn deep contextual representations and have achieved state-of-the-art results on many NLP tasks. Kaliyar et al. [11] proposed FakeBERT, which combines BERT embeddings with parallel one-dimensional convolution blocks for fake news classification. Based on this survey, the proposed work compares content-based classical, hybrid deep learning and transformer models on the same datasets.

III. PROPOSED METHODOLOGY

A. Datasets

Two public datasets are used: the ISOT Fake News dataset [5], which contains full articles labelled as real or fake, and the LIAR dataset [3], which contains short

statements. For LIAR, the six labels are mapped into two classes (true, mostly-true and half-true as real; barely-true, false and pants-fire as fake). Each dataset is split into 80% training, 10% validation and 10% test data using stratified sampling.

B. Preprocessing and Feature Extraction

The headline and body are concatenated to form the input text. The preprocessing steps include removal of URLs, HTML tags, special characters and extra spaces, lower-casing, tokenisation, stop-word removal and lemmatisation. Source names and agency tags such as “(Reuters)” are removed so that the model cannot learn a shortcut from the publisher name. For classical models, TF-IDF [12] vectors of unigrams and bigrams are extracted. For deep learning models, the text is converted to integer sequences and padded to a fixed length.

C. Classical Machine Learning Models

Logistic Regression, Multinomial Naive Bayes, Linear Support Vector Machine and Random Forest classifiers are trained on TF-IDF features. Hyper-parameters are selected using grid search with 5-fold cross-validation on the training set.

D. Hybrid CNN–LSTM Model

The input sequence is passed through an embedding layer initialised with pre-trained GloVe vectors. A one-dimensional convolution layer with max-pooling extracts local phrase features, and an LSTM layer then models the order of these features across the article. The output is passed through a dropout layer and a dense layer with sigmoid activation to predict the probability of the article being fake.

E. BERT-Based Model

The pre-trained bert-base-uncased model [10] is fine-tuned for binary classification. The article is tokenised with the WordPiece tokenizer and truncated to 512 tokens. The [CLS] representation is fed to a classification head, and the whole model is trained using the AdamW optimiser and binary cross-entropy loss.

F. Evaluation

All models are compared on the same test sets using accuracy, precision, recall, F1-score and the ROC-AUC. Since wrongly marking fake news as real is more harmful, recall of the fake class is given special importance.

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